Mathematics III – Trigonometry

B.Tech study material for AI Personalized Tutor RAG

1. Trigonometric Ratios

• sin θ = opposite / hypotenuse

• cos θ = adjacent / hypotenuse

• tan θ = opposite / adjacent

• cosec θ = hypotenuse / opposite

• sec θ = hypotenuse / adjacent

• cot θ = adjacent / opposite

2. Fundamental Identities

• $\sin^2\theta + \cos^2\theta = 1$

• $1 + \tan^2\theta = \sec^2\theta$

• $1 + \cot^2\theta = \operatorname{cosec}^2\theta$

• tan θ = sin θ / cos θ

• cot θ = cos θ / sin θ

• sec θ = 1 / cos θ

• cosec θ = 1 / sin θ

3. Standard Values

• Angles: 0°, 30°, 45°, 60°, 90°

• sin θ: 0, 1/2, 1/√2, √3/2, 1

• cos θ: 1, √3/2, 1/√2, 1/2, 0

• tan θ: 0, 1/√3, 1, √3, undefined

4. Quadrants and Signs

• Quadrant I: all six functions are positive.

• Quadrant II: sine and cosecant are positive.

• Quadrant III: tangent and cotangent are positive.

• Quadrant IV: cosine and secant are positive.

5. Allied Angles

• sin(90° – θ) = cos θ

• cos(90° – θ) = sin θ

• tan(90° – θ) = cot θ

• sin(180° – θ) = sin θ

• cos(180° – θ) = –cos θ

• tan(180° – θ) = –tan θ

• sin(180° + θ) = –sin θ

• cos(180° + θ) = –cos θ

• tan(180° + θ) = tan θ

• $\sin(360° - \theta) = -\sin\theta$

• $\cos(360° - \theta) = \cos\theta$

• $\tan(360° - \theta) = -\tan\theta$

6. Compound Angles

• $\sin(A+B) = \sin A \cos B + \cos A \sin B$

• $\sin(A-B) = \sin A \cos B - \cos A \sin B$

• $\cos(A+B) = \cos A \cos B - \sin A \sin B$

• $\cos(A-B) = \cos A \cos B + \sin A \sin B$

• $\tan(A+B) = (\tan A + \tan B)/(1 - \tan A \tan B)$

• $\tan(A-B) = (\tan A - \tan B)/(1 + \tan A \tan B)$

7. Double and Triple Angles

• $\sin 2A = 2 \sin A \cos A$

• $\cos 2A = \cos^2 A - \sin^2 A = 2\cos^2 A - 1 = 1 - 2\sin^2 A$

• $\tan 2A = 2\tan A/(1 - \tan^2 A)$

• $\sin 3A = 3\sin A - 4\sin^3 A$

• $\cos 3A = 4\cos^3 A - 3\cos A$

8. Half-Angle Formulas

• $\sin^2(A/2) = (1 - \cos A)/2$

• $\cos^2(A/2) = (1 + \cos A)/2$

• $\tan(A/2) = \sin A/(1 + \cos A)$

• $\tan(A/2) = (1 - \cos A)/\sin A$

9. Product-to-Sum

• $2 \sin A \cos B = \sin(A+B) + \sin(A-B)$

• $2 \cos A \cos B = \cos(A+B) + \cos(A-B)$

• $2 \sin A \sin B = \cos(A-B) - \cos(A+B)$

10. Solving Trigonometric Equations

• If $\sin\theta = 1/2$, then $\theta = 30° + 360°n$ or $150° + 360°n$.

• If $\cos\theta = 1/2$, then $\theta = 360°n \pm 60°$.

• If $\tan\theta = 1$, then $\theta = 45° + 180°n$.

• Always check the specified interval and the quadrants where the function has the required sign.

11. Common B.Tech Problem Types

• Evaluate expressions using standard values.

• Prove identities using fundamental identities.

• Find an unknown ratio from a known ratio and quadrant.

• Apply compound-angle formulas.

• Apply double-angle and half-angle formulas.

• Solve trigonometric equations on a specified interval.

• Determine signs in different quadrants.

12. Worked Examples

• Example 1: If $\sin\theta = 3/5$ and θ is acute, $\cos^2\theta = 1 - 9/25 = 16/25$, so $\cos\theta = 4/5$.

• Example 2: If $\tan A = 1/2$, $\tan 2A = 2(1/2)/(1 - 1/4) = 4/3$.

• Example 3: $\sin\theta = 1/2$ on $0° \le \theta \le 360°$ gives $\theta = 30°, 150°$.

13. Recommended Learning Sequence

• Learn the six basic ratios.

• Learn fundamental identities.

• Practice standard-angle values and quadrant signs.

• Study allied angles.

• Study compound angles.

• Study double, triple, and half angles.

• Practice identity proofs and simplification.

• Finish with mixed trigonometric equations.

14. RAG Retrieval Keywords

• trigonometric ratios; identities; standard values; quadrants; allied angles; compound angles; double angles; triple angles; half angles; product-to-sum; trigonometric equations; identity proof; simplification; sin; cos; tan; sec; cosec; cot.